resilience to diseases does not have the multiplier effect in India, as it did in Brazil for 50 years. While a few of our top performing cattle can produce up to 20 litres of milk a day, our average remains 8-10 litres." Naturally, genetic improvement through artificial insemination is more widespread in Brazil. According to Gujarat animal husbandry department director Dr Falguni Thakar, even artificial intelligence is used to govern about 40 per cent of cattle breeding.

The real spur for the import of high-quality Gir bull semen to bolster India's milk production is a projected increase of consumption. India's estimated consumption of milk/ milk products was 320 million litres a day in 2019 and it is projected to grow to 468 ML by 2030.

Production of milk can be increased through an increase in the number of cattle or increase in the productivity of the existing cattle. A large increase in numbers is insupportable—shrinking grazing land would drive stray cattle on to roads and farmlands. The second option can be achieved by better feeding/ management of existing cattle, but it cannot give results beyond the animal's genetic makeup. This was why a largescale long-term genetic improvement programme—involving artificial insemination methods—was initiated by the NDDB in 2008.

Artificial insemination through frozen semen is a widely used technique where farmers approach a semen station to get their cow inseminated. Typically, semen of bulls with superior genetic makeup—with progeny proven to have given higher milk output—is used. When inseminated with sex-sorted semen, there is a 90 per cent probability of birthing a female. With ever-increasing use of mechanised vehicles like tractors, the utility of bulls has fallen hugely. That way, sex-sorted semen resolves the issue of stray cattle.

Brazil now has four million purebred Gir cows; India has only 2.3 million. It has preserved the breed better than India



Gir was not the only Indian breed cow to be sent to Brazil. The milch breed Red Sindhi, and draught breeds like Ongol and Nellore from Andhra Pradesh and Kankrej from Gujarat, have been bred in Brazil for long. It just so happens that Gir cows have better resistance to diseases and have higher average milk production than other breeds.

Since the early '60s, India's genetic improvement programme of cattle focused on the cross-breeding of such high milk-producing breeds like Holstein Friesian and Jersey with Indian cattle. Presently, crossbreeds contribute one-third of India's bovine milk, whereas their population is 22 per cent of the total bovine population. Historically, this is what helped India achieve self-sufficiency in milk production. "The crossbred cattle will continue to contribute a significant share of milk production. Current genetic improvement programmes, apart from focusing on pure indigenous breeds, also seek to improve crossbred cattle," a source at NDDB says. In recent years, the programme imported high-quality bulls and frozen embryos of quality breeds.

This project has courted controversy over the years. In 2019, RSS chief Mohan Bhagwat questioned the government's decision to float a tender for the import of 100,000 Gir bull semen. Saurashtra farmers had also raised concerns that the imported semen would be of other breeds crossbred with Gir cows and inseminating our indigenous cows with it would end up ruining the genetic pool, with the resultant progeny being too weak to survive Indian conditions. However, NDDB scientists have clarified that only semen of purebred Gir bulls is being imported. ■